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What Does a System Administrator Do? Your Career Guide

Written by Coursera Staff • Updated on Jul 23, 2026
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System administrators support, troubleshoot, and maintain computer servers and networks.



Key takeaways

- A system administrator is responsible for installing, configuring, troubleshooting, upgrading, and maintaining an organization’s software and hardware equipment.
- Help desk technicians, network analysts, and computer systems analysts are examples of entry-level system administrator jobs.
- To become a system administrator, you’ll typically need a bachelor’s degree, specialized information technology (IT) skills, and relevant certifications that qualify your expertise.

As a system administrator, you can make $96,800 annually in the United States [1]. Explore the role of system administrators, including their salary prospects, technical skills, and popular certifications.

Afterward, if you're ready to build essential skills to succeed as a system administrator, enroll in IBM's Information Technology (IT) and Cloud Fundamentals Specialization. Beginner-friendly, this program offers guidance on operating systems, commonly used storage and network types, hardware connectors, database systems, and more.



specialization

Information Technology (IT) and Cloud Fundamentals

Kickstart a new IT and Cloud Career . Master hardware, operating systems, programming, database, networking, storage, cybersecurity, and Cloud computing essentials.

4.7

(12,204 ratings)

21,499 already enrolled

Beginner level

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Average time: 2 month(s)

Learn at your own pace

Skills you'll build:

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Systems, Linux, Mac OS, Wireless Networks, Network Protocols, Local Area Networks, Computer Networking, Command-Line Interface, Networking Hardware, Network Model, Wide Area Networks, Microsoft Windows, Network Planning And Design, Network Security, Network Architecture, Network Administration, OSI Models, TCP/IP

What is a system administrator?

System administrators, also known as sysadmins, are IT professionals who ensure an organization's computer systems function properly and meet business requirements. Sysadmins support, troubleshoot, and maintain computer servers and networks.

What does a system administrator do?

A system administrator job description might include:

- Managing Windows, Linux, or Mac systems
- Upgrading, installing, and configuring application software and computer hardware
- Troubleshooting and providing technical support to employees
- Creating and managing system permissions and user accounts
- Performing regular security tests and security monitoring
- Maintaining networks and network file systems

System administrator salary and job outlook

A system administrator's average base salary in the US is $96,800, according to the US Bureau of Labor Statistics (BLS) [1]. Salaries can vary depending on experience and the cost of living where you're located. The bureau forecasts an average of 14,300 annual job openings every year for network and computer systems administrators between 2024 and 2034 [1].

Read more: System Administrator Salary: Guide

System administration vs. network administration

Some organizations might use system administrator and network administrator interchangeably due to the many overlapping responsibilities, but these roles have their differences. System administration focuses on servers and computer systems, while network administrators work more specifically with network-related tasks and equipment, like setting up routing, IP addresses, and maintaining local area networks (LANs).

If you're in a smaller organization, these responsibilities might be folded into one role, while larger organizations may tend to differentiate them. Since systems and networks are often intertwined, it's not rare to see job descriptions that require knowledge of both.

How to become a system administrator

Becoming a system administrator will require learning some specialized IT skills. If you're new to the field, you might consider one of the following system administrator career path roles below. If you're already an IT professional, earning a certification or taking online courses can provide experience with industry-standard tools and showcase your skills to potential employers.

Already have some experience? Sharpen your system administration skill set and prepare to earn an industry-recognized Linux certificate with the Linux Foundation Certified System Administrator (LFCS) Specialization. In as little as eight weeks, you can master the key concepts on the LFCS exam while practicing real-life Linux administration tasks.



specialization

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Prepare to Pass the LFCS Certification Exam.. Learn core Linux concepts and become familiar with the objectives of the LFCS certification exam.

4.6

(17 ratings)

2,581 already enrolled

Intermediate level

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Average time: 2 month(s)

Learn at your own pace

Skills you'll build:

Lightweight Directory Access Protocols, Virtual Machines, OS Process Management, Remote Access Systems, Containerization, Linux, Linux Commands, Git (Version Control System), Ubuntu, File Systems, Firewall, Shell Script, Operating System Administration, Virtualization, Systems Administration, Linux Administration, Virtualization and Virtual Machines, Bash (Scripting Language), File Management, Package and Software Management, Network Administration, System Configuration, Virtual Networking, Software Installation, Command-Line Interface, Open Source Technology, Authentications, Service Management, System Monitoring, General Networking, User Accounts, Authorization (Computing), Data Storage, Performance Tuning, Network Security, Security Controls, Hardening, Data Storage Technologies

System administrator jobs: Entry-level feeder roles

If you're transitioning into IT from a related career or considering starting down the path to system administration, these jobs can serve as stepping stones to your goal and prepare you for success in your first role as a system administrator.

- Help desk technician
- IT support specialist
- Network analyst
- Computer systems analyst

System administrator skills

As a sysadmin, you'll need to develop a core set of skills that will allow you to make sure the computer systems in your organization are running smoothly. Here are the top skills that you'll want on your resume:

- **In-depth knowledge of** operating systems**:** Whether it's Windows, Linux, or Mac, hiring managers generally look for competency in administering the operating system used in their company. Windows and Linux operating systems have been widely embraced and are good systems to start with, though it's a good idea to check if your industry favors one over the other.
- **Familiarity with hardware:** Working with physical devices such as servers or printers is an important part of a system administrator's job.
- Cloud computing skills**:** Familiarity with cloud applications like Microsoft 365, Google Cloud Platform, and Amazon Web Services (AWS) can boost your chances of being hired.
- **Knowledge of networks:** System administrators are often expected to be able to set up and maintain LANs and wide area networks (WANs), as well as network security features like firewalls. This might often be the case at smaller companies, where IT professionals are expected to wear many hats.
- **Communication and** interpersonal skills**:** In addition to working on a team, sysadmins are often expected to help other employees who don't have the same technical knowledge through help desk support or other means. Being able to communicate well will be a key part of a system administrator's successful day-to-day tasks.

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Certifications for system administrators

There are several certifications you can take in order to gain experience as a systems administrator.

Windows Server Administration Fundamentals

Having knowledge of Windows servers is likely to be a significant asset as a system administrator. The Windows Server Administration Fundamentals certification will qualify your expertise in Windows servers, networking, Active Directory, account management, and system recovery tools.



course

Windows Server Management and Security

Microsoft Windows has been at the forefront of enterprise computing for several decades. What most office workers see is the desktop side – such as Windows ...

4.7

(682 ratings)

30,698 already enrolled

Beginner level

Learn More

Average time: 12 hour(s)

Learn at your own pace

Skills you'll build:

Microsoft Windows, Active Directory, System Configuration, Authorization (Computing), Encryption, Operating System Administration, Computer Networking, Servers, Microsoft Servers, Identity and Access Management,

Enterprise Security, Windows Servers, User Accounts, Network Performance Management, Group Policy, Software Installation, Authentications, Security Controls

Google IT Support Professional Certificate

The Google IT Support Professional Certificate equips you with the skills to tackle common sysadmin tasks such as customer support and troubleshooting. It also introduces you to commonly used systems like Linux and Domain Name Systems, and prepares you for the CompTIA A+ exam. The course will earn you an industry-recognized professional certificate for your studies and requires no previous experience.



professional certificate

Google IT Support

The launchpad to a career in IT. This program is designed to take beginner learners to job readiness in about three-to-six months.

4.8

(216,071 ratings)

2,208,407 already enrolled

Beginner level

Learn More

Average time: 3 month(s)

Learn at your own pace

Skills you'll build:

Chef (Configuration Management Tool), IT Infrastructure, Computer Networking, Ruby (Programming Language), Version Control, Network Administration, Operating System Administration, Information Systems Security, IT Security Architecture, Package and Software Management, Interviewing Skills, Systems Administration, Git (Version Control System), Network Security, TCP/IP, IT Automation, General Networking, Web Presence, Computer Hardware, Network Troubleshooting, Linux Commands, File Systems, Software Installation, Remote Access Systems, System Monitoring, Operating Systems, User Accounts, OS Process Management, File Management, Microsoft Windows, Command-Line Interface, Linux, System Support, Authorization (Computing), Linux Administration, Cloud Services, Disaster Recovery, Active Directory, Lightweight Directory Access Protocols, Server Administration, Servers, Network Infrastructure, Technical Consulting, Cloud Management, Cloud Infrastructure, Generative AI, Branding, AI literacy, Google Gemini, Encryption, Malware Protection, Network Monitoring, Security Strategy, Hardening, Cyber Security Policies,

Security Awareness, Firewall, Cyber Attacks, Authentications, Cryptography, Data Security, Application Security, Computer Security Awareness Training, Cybersecurity, Identity and Access Management, Network Protocols, Network Model, Wireless Networks, Data Integrity, Network Routing, Dynamic Host Configuration Protocol (DHCP), Routing Protocols, Networking Hardware, Technical Support, Information Technology, Technical Documentation, Technical Writing, Technical Communication, Customer Support

Red Hat System Administrator Certification

Getting the Red Hat System Administrator Certification will give you the experience with Linux operating systems you need to land an entry-level job in systems administration.



course

Fundamentals of Red Hat Enterprise Linux

This course will provide you with a basic introduction to Linux® skills using Red Hat® Enterprise Linux 9. It will show you how a Linux system is organized ...

4.8

(113 ratings)

10,766 already enrolled

Beginner level

Learn More

Average time: 10 hour(s)

Learn at your own pace

Skills you'll build:

System Monitoring, Open Source Technology, Linux Commands, User Accounts, Package and Software Management, Network Administration, OS Process Management, Command-Line Interface, File Management, Linux Administration, Red Hat Enterprise Linux, Operating System Administration, General Networking, Remote Access Systems, Linux

CompTIA Server+ or A+

CompTIA offers several certifications that are designed to prepare you for entry-level positions. The Server+ or A+ certifications in particular will leave you with the necessary skills and experience under your belt to start

applying for entry-level IT jobs. While you'll gain specialized knowledge around servers with the Server+ certification, the A+ certification covers a broader set of skills applicable to other aspects of IT.

Preparing for the sysadmin interview

Preparing for any interview means anticipating some questions ahead of time and knowing how to answer them. Here are some basic questions you might encounter when you interview for a system administrator role.

- Describe your experience with hardware.
- How would you deal with a virus?
- How would you know when to upgrade a server?
- Tell us about a time when you weren't able to solve an issue. What did you do?
- What qualities do you have that make you a good system administrator?

Interviewing for IT jobs

Since IT jobs are highly technical, interviewers are likely to ask you several technical questions in addition to more personal behavioral questions. Be prepared for both. Refresh your IT skills by mentally walking through several tasks you know how to perform, and practice explaining them aloud.

It's a good idea to think about some stories of when something went well for you, when things didn't go so well, and what you did in that situation to make it better. Having concrete examples at the ready can help you feel mentally prepared and appear more confident to employers.



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Frequently asked questions (FAQ)

What are the four types of administrators?
Does system administration require coding?
Can I become a system administrator without a degree?

Article sources

1.

US Bureau of Labor Statistics. "Network and Computer Systems Administrators, https://www.bls.gov/ooh/computer-and-information-technology/network-and-computer-systems-administrators.htm." Accessed April 7, 2025.

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Skills

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- Cybersecurity
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- Microsoft Excel
- Project Management
- Python
- SQL

Professional Certificates

- Google AI Certificate
- Google Cybersecurity Certificate
- Google Data Analytics Certificate
- Google IT Support Certificate
- Google Project Management Certificate
- Google UX Design Certificate
- IBM AI Engineering Certificate
- IBM AI Product Manager Certificate
- IBM Data Science Certificate
- Intuit Academy Bookkeeping Certificate

Courses & Specializations

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- AI in Healthcare Specialization
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Career Resources

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